

Map-STDP

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Abstract

STDP is a biologically plausible learning rule that has been widely used in spiking neural networks. However, STDP is a local learning rule that cannot capture the global modular structure of the neural network. In this paper, a new Map-STDP rule is derived using the map equation, which is an info-theoretic objective function for community detection.

1 Introduction

Spike-timing dependent plasticity (STDP) is a biologically plausible learning rule that has been widely used in spiking neural networks. When a pre-synaptic neuron fires shortly before a post-synaptic neuron, the synapse between them is strengthened. Conversely, if the pre-synaptic neuron fires shortly after the post-synaptic neuron, the synapse is weakened.

However, STDP is a fully local rule that cannot capture the global structure of the neural network. This issue is much more prominent when population vectors are used for inference via the population of neurons in different regions of the network. In order to ensure the expressivity of the population vectors, the neural network must have a modular structure, much like the cortical organization of the brain.

Moreover, the brain has been identified to perform both stochastic random walks and entropy maximization [1], which are two major components of the map equation, an info-theoretic objective function for community detection in the Infomap algorithm. The map equation measures the modular structure of a network by quantifying the amount of information needed to describe the flow of activity through the network.

As such, a Map-STDP rule that incorporates the map equation is derived. The Map-STDP rule is a local learning rule that can capture the global modular structure of the neural network. Moreover, the Map-STDP rule preserves biorealism through

the use of info-theoretic entropy maximization, stochastic neural dynamics, and local synaptic updates.

2 Results

3 Cross referencing

4 Examples for theorem like environments

5 Conclusion

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

References

- [1] Dayan, P., Abbott, L.F.: Information theory. In: Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems, pp. 123–150. MIT Press, Cambridge, MA (2001). <https://doi.org/10.7551/mitpress/6755.001.0001>